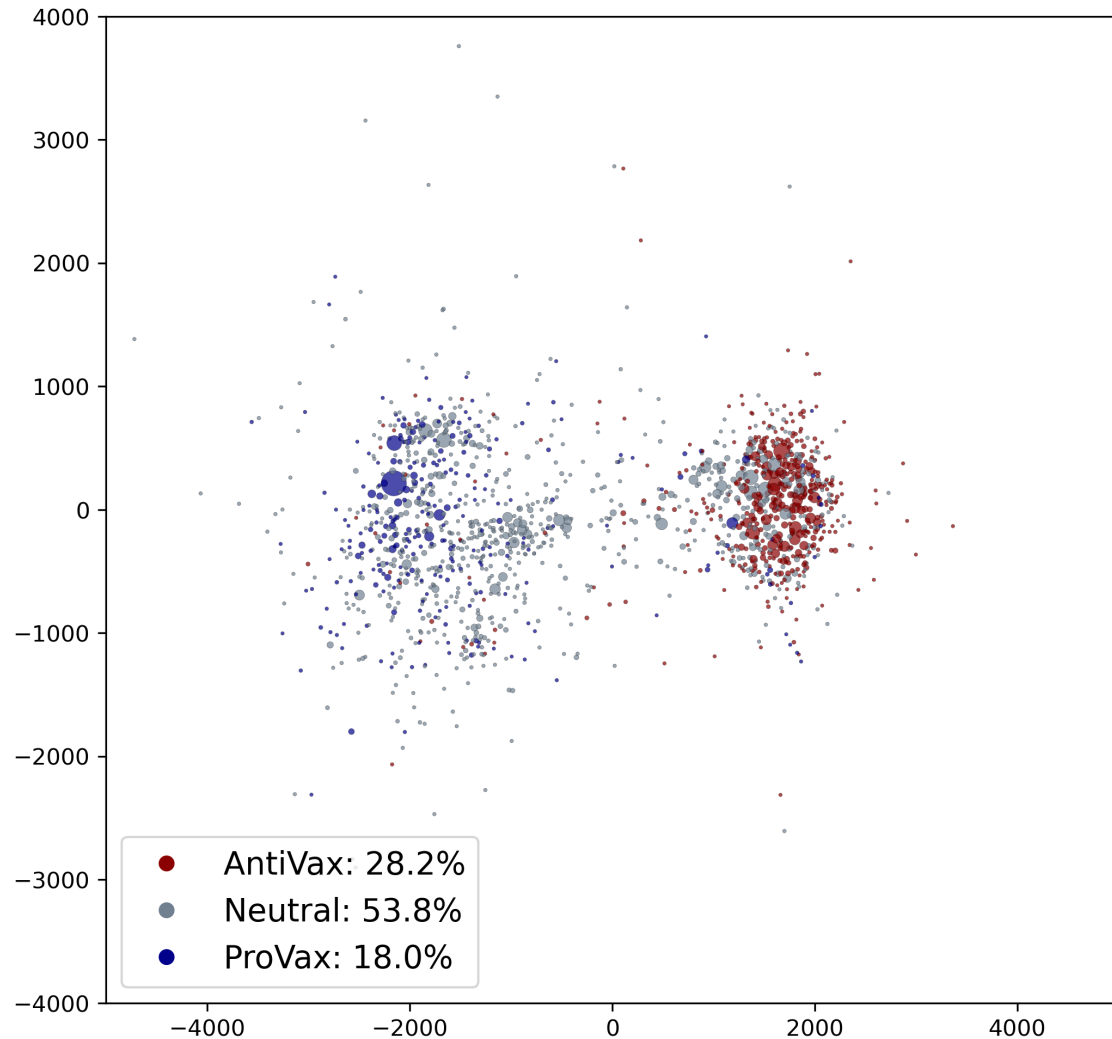


Twitter retweet network about «vaccine» & other topics



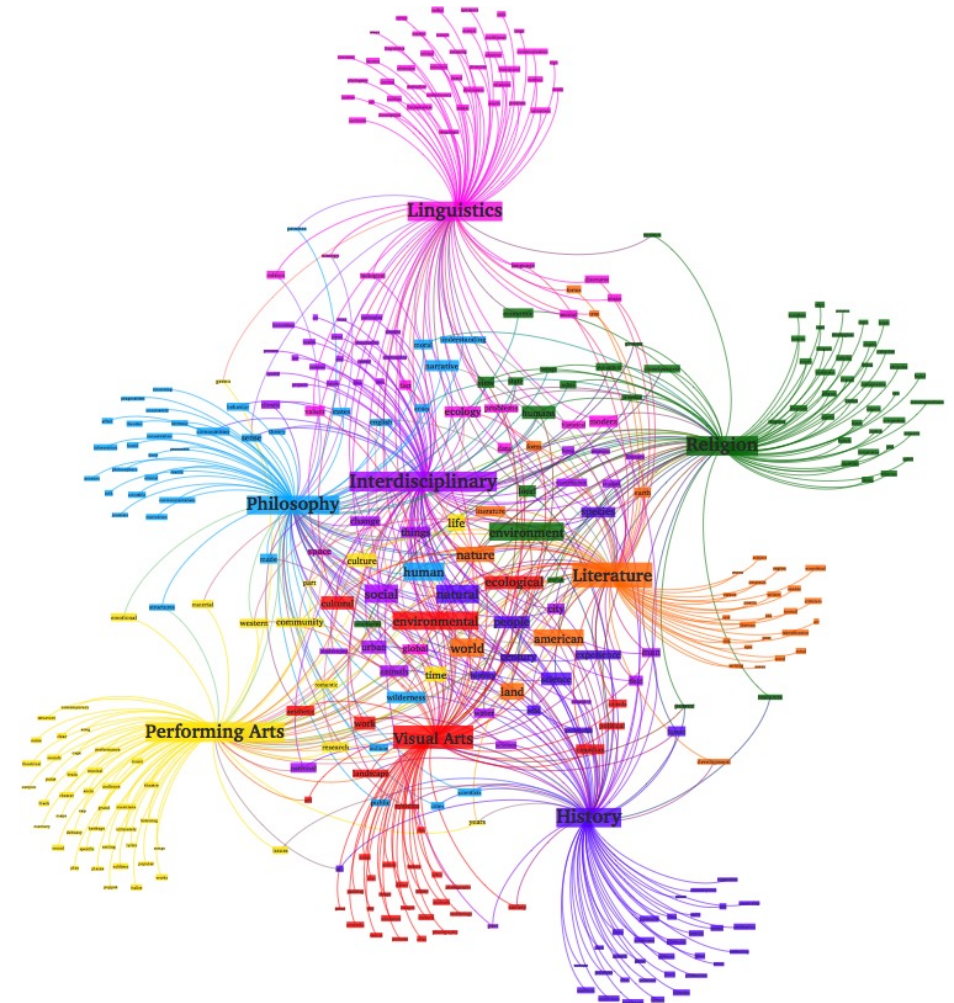
User/tweet classification, link prediction:
graph neural networks

Multiplex: analysis with same users
different topics. role of users, communities,
centralities, node embedding and link
prediction

Discriminate features of nodes from
different/opposite communities: why are
they different?

Scientific literature automated search and analysis

- Collect scientific abstracts from online repositories through automated calls to Application Programming Interfaces (Scopus, PubMed)
- Topic modelling: transform abstracts into vectors and clusterize them to extract the different topics (*extendible to Pattern Recognition*)
- Build WordNets between words
- Build and analyze (co-)citation networks between scientific authors (pagerank-like)
- Guess citation number
- Study network evolution



Web source: <https://datasciencelady.com/2020/09/10/latest-topic-modelling-research-papers-a-quick-overview/>

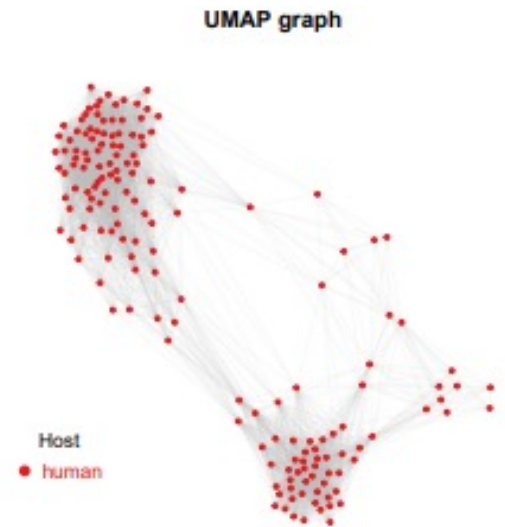
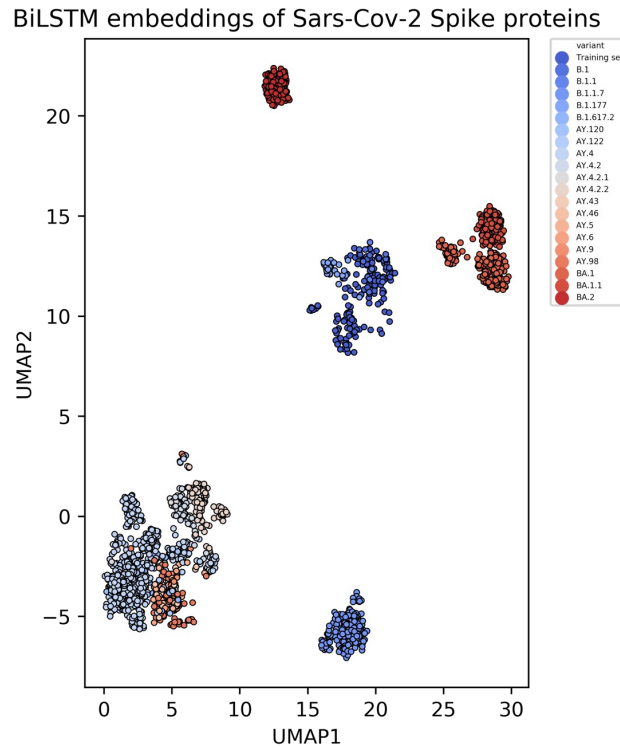
Social & text network projects

- Characterize multiplexes (same users different topics)
- Evolution over time (timed tweets or publications & citations)
- Evolution over time of node classification ("gray zone")
- Ranking paper importance by citation growth
- Identify paper communities & understand meaning (title, abstracts)
- Evolution over time of retweet/citation/coauthorship networks

Language models for protein sequences

Embed protein sequences to vectors («from strings to numbers»: word2vec, transformers, LSTM)

- Build networks based on proximity measures and compare them with biologic metadata (Sars-CoV-2 variants, influenza databases)
- Explore the effect of mutations in the embedding space
- Generation of synthetic data at the antigenic map level (data imputation)



Left) UMAP 2d dimensionality reduction of 1024 dimensional embeddings of Sar-CoV-2 sequences.

Right) Network built on similarity measures between pairs of protein sequences in the embedding space.

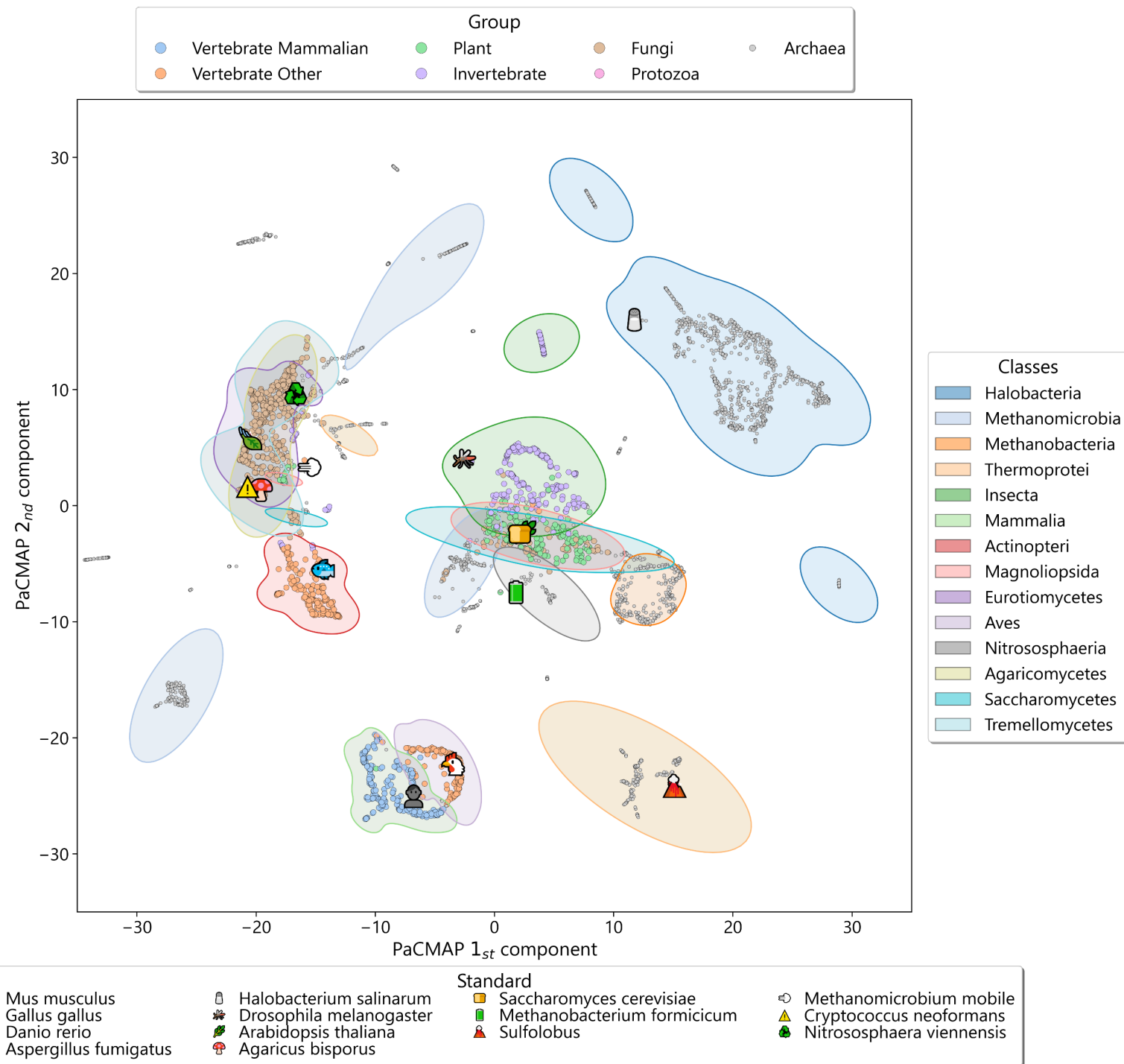
Evolutionary biology: "systems" approach

Map organism proteins in vector space (e.g. with known DL algorithms)

Find similarities between organisms (functional classes), guess protein functionality

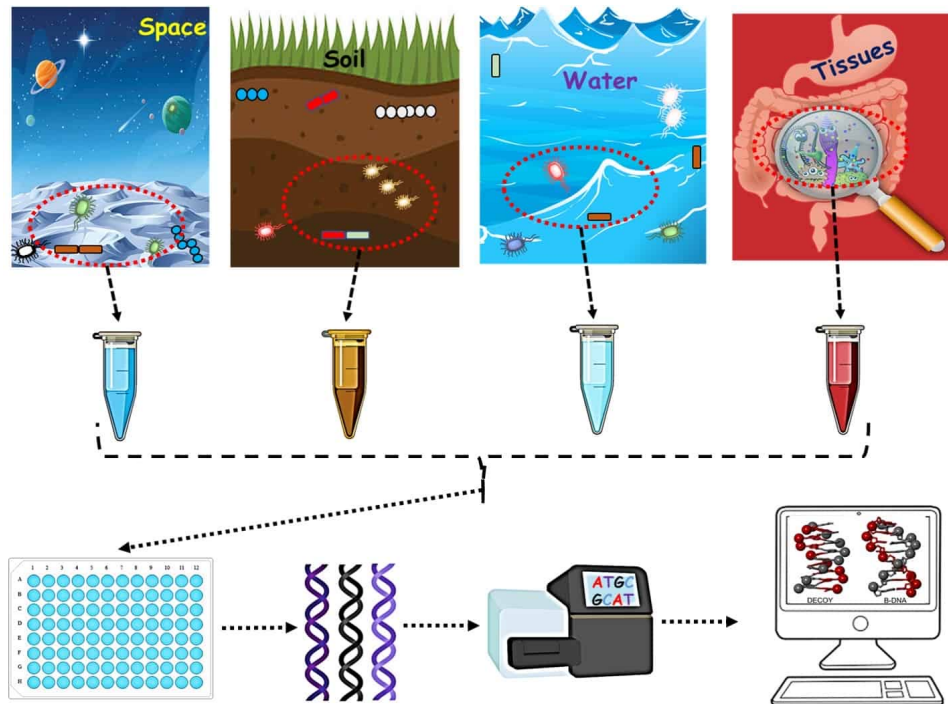
Map organisms in a vector space "functional phylogeny"

Link prediction between organisms (considering organism-specific PPIs available)

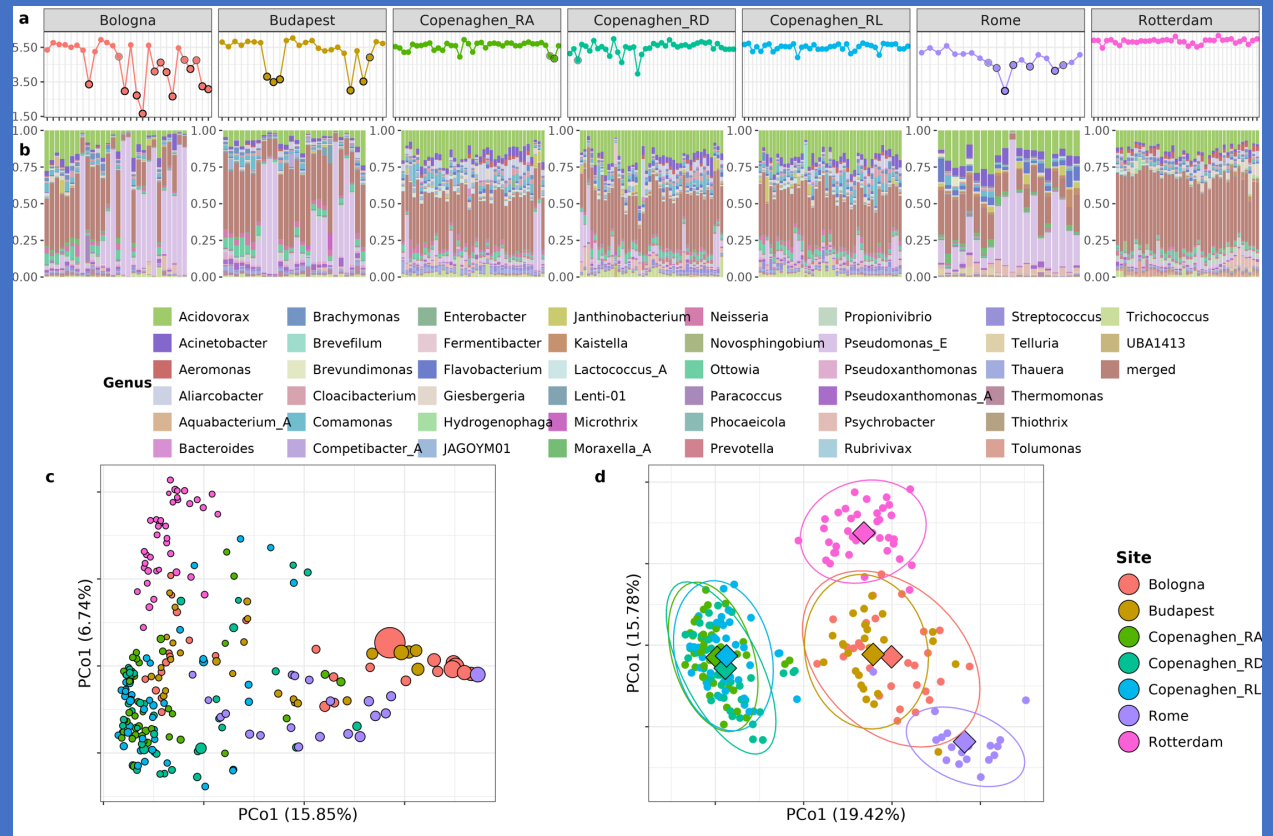
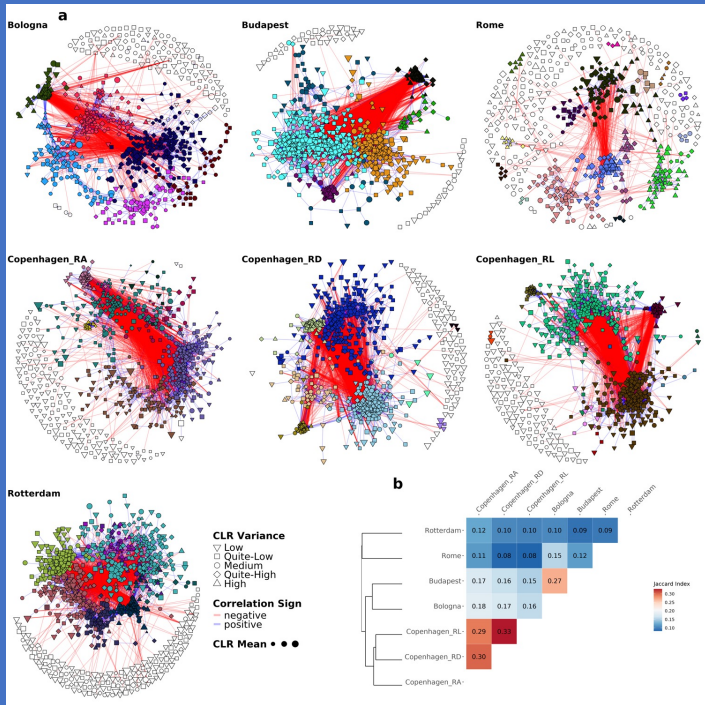


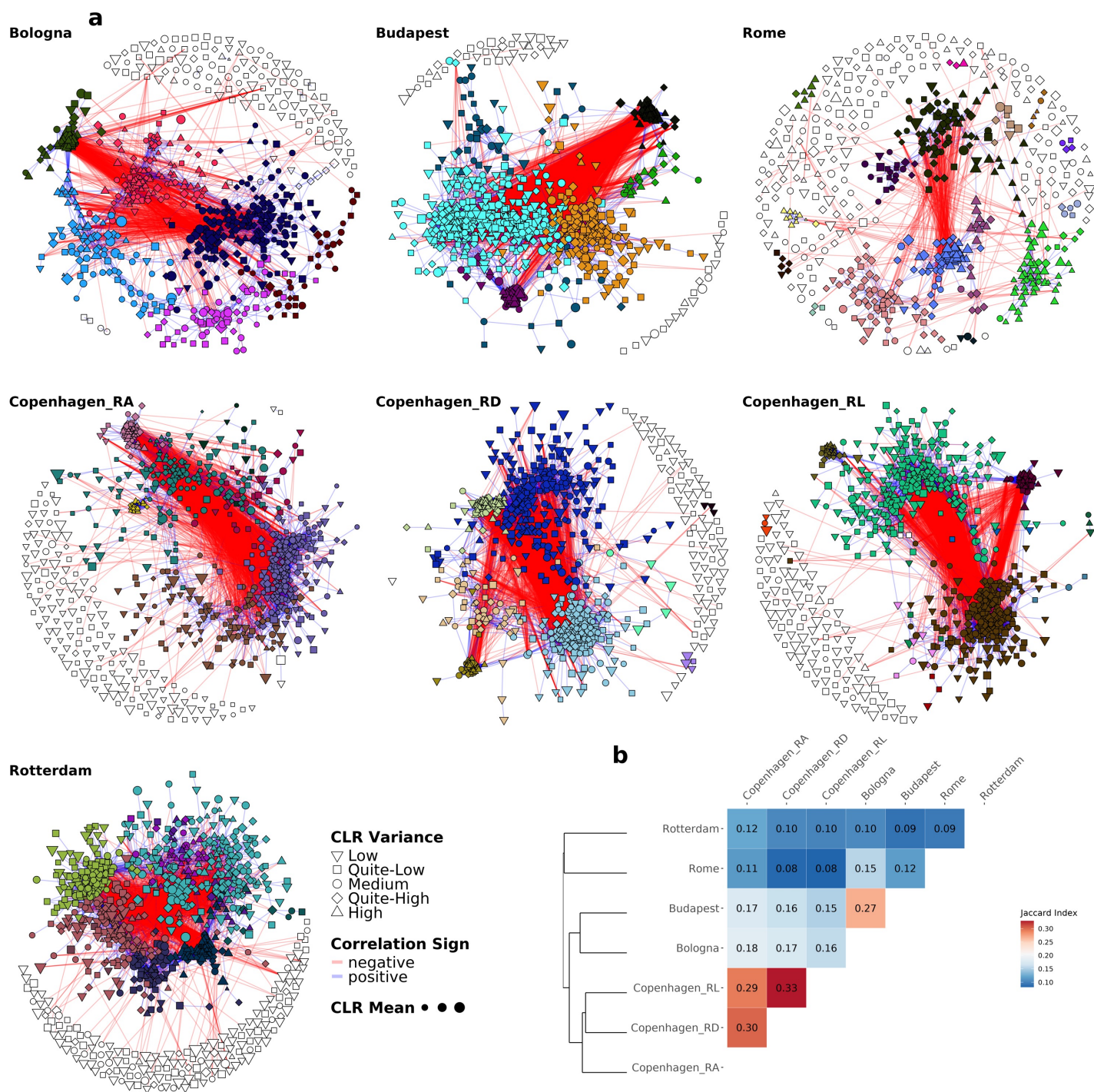
Metagenomic Network analysis

Study of the ecological systems produced by microorganisms in their natural environment:
network communities (signed graphs correlation/anticorrelation)
high-order relations (e.g. triadic correlation/association)



	OTU_185	OTU_108	OTU_160	OTU_70	OTU_13	OTU_186	OTU_39	OTU_3	OTU_1	OTU_128	OTU_22
69-001-1010	4	0	0	242	0	0	39	982	10820	0	289
69-001-1011	0	2	0	71	0	0	74	1131	11302	0	706
69-001-1012	0	11	0	67	0	0	168	256	3058	0	323
69-001-1015	0	11	0	301	0	11	37	2050	12942	22	143
69-001-1021	1	37	0	91	0	0	73	369	5200	6	321
69-001-1022	0	84	0	107	0	0	629	389	4564	132	609
69-001-1022.1	0	17	0	73	0	1	27	263	4156	39	525
69-001-1023	0	24	0	50	0	0	86	515	4321	31	266
69-001-1024.1	0	89	0	117	0	0	28	422	3987	4	685
69-001-1025	1	98	0	133	0	0	41	889	3338	3	561
69-001-1042	0	51	0	150	3	0	57	319	2446	1	803
69-001-1043	5	39	0	232	53	0	108	212	2284	1	742
69-001-2001	0	3	0	128	0	0	10	222	9713	27	44
69-001-2002	0	6	0	108	0	0	34	813	6328	25	215
69-001-4021	0	21	0	42	0	0	278	847	3405	12	215
69-001-4021.18	0	66	0	53	0	0	46	339	2838	25	244
69-001-4023	4	32	0	58	0	0	103	509	2185	3	553
69-001-4023.1	2	78	0	149	0	0	191	699	3091	37	1051
69-001-4023.3	0	53	0	159	0	0	120	338	2483	27	765
69-001-4024	0	103	0	76	0	0	131	595	2207	91	543
69-001-4024.1	0	25	0	64	0	0	91	311	2540	15	380
69-001-4025	0	176	0	68	0	0	434	306	5728	258	865
69-001-4025.1	1	11	0	33	0	0	130	380	1914	21	407





Network across different Sites

- Vertex are the bacterial species
- Links are significant correlations calculated on the CLR transformed abundances selected making a threshold on adjusted p-value though false discovery rate set to 0.01
- Vertex color are associated to communities (signed modularity)

Edges Jaccard index indicates strong similarities between the three sites in Copenhagen and between Bologna and Budapest.

3D structure of DNA and proteins

DNA

Misfolded DNA



Cancer

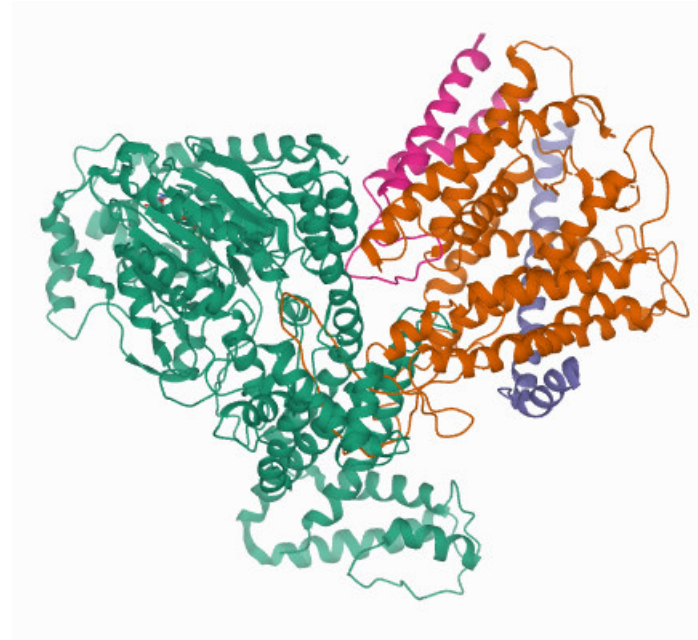


Protein

Misfolded
protein

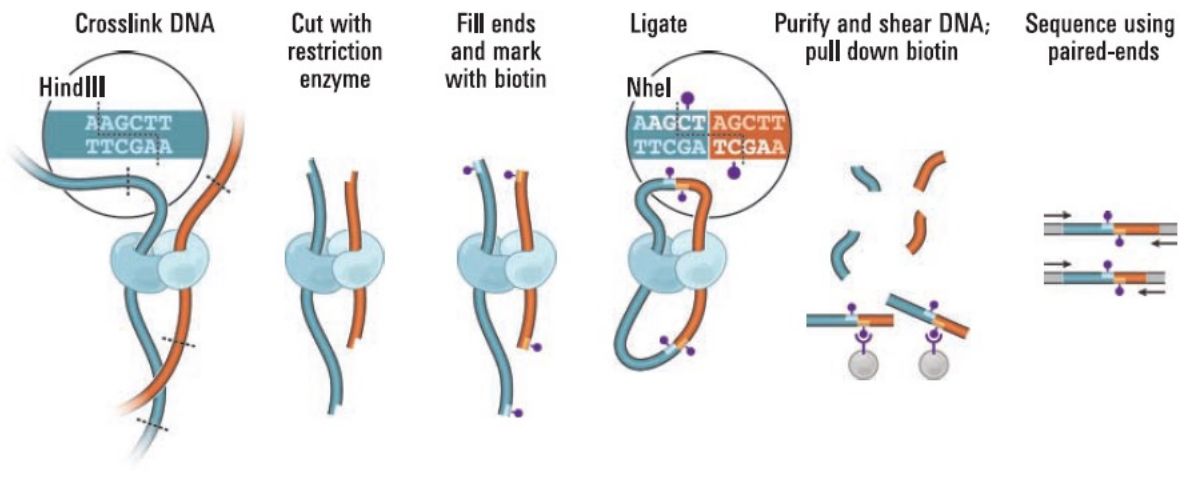


diseases



The **3D structure** of DNA and proteins determines their **biological functions**

Studying translocations using Hi-C data

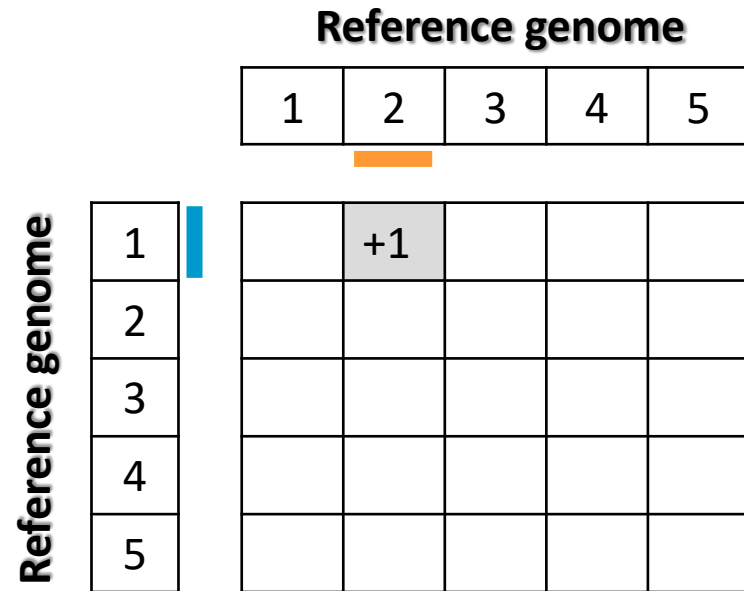


		Reference genome				
		1	2	3	4	5
Reference genome	1		+1			
	2					
	3					
	4					
	5					

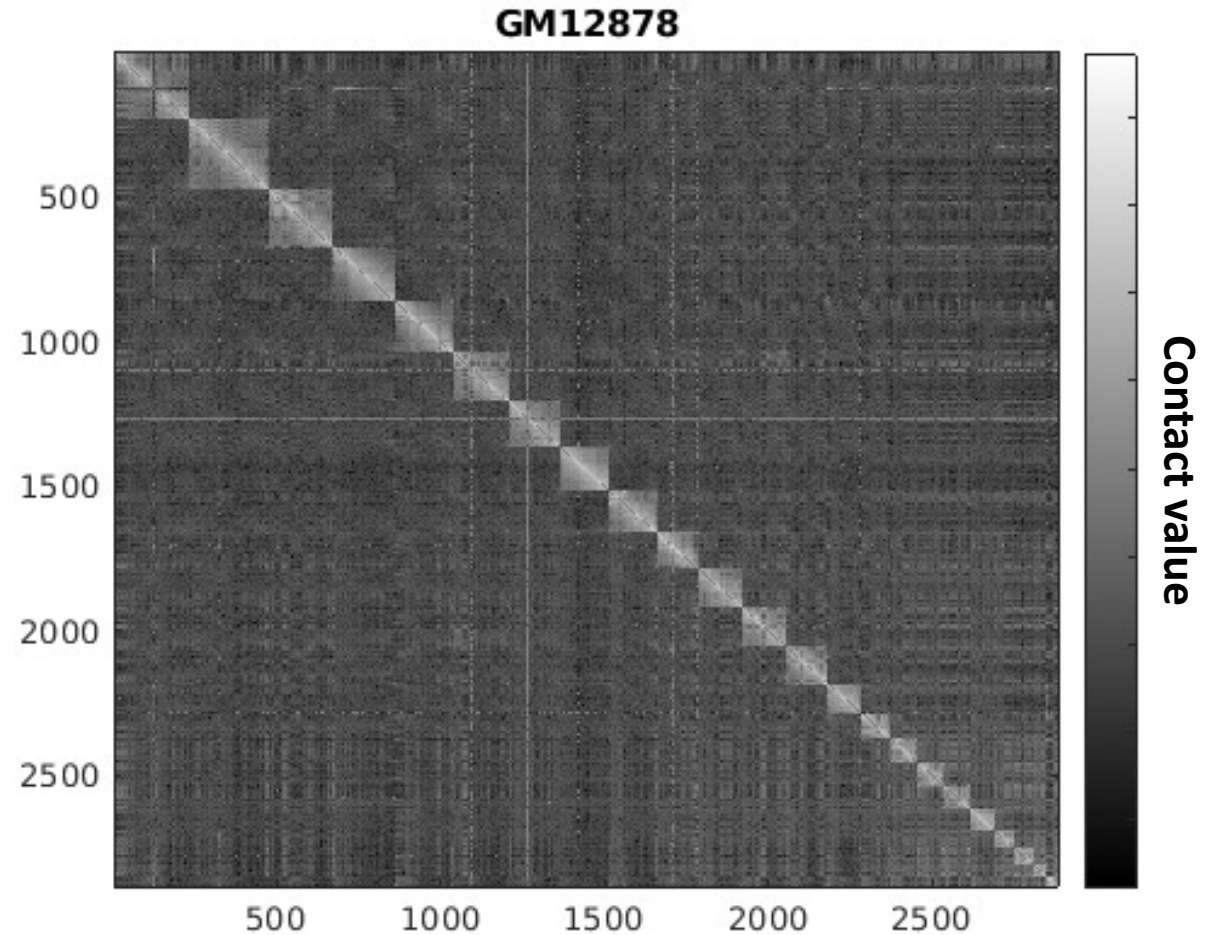
Overview of Hi-C experimental technique.



Hi-C contact map: weighted (almost) completely connected graph



The **higher the contact value**, the closer the bins in the nuclear space.



Projects

Protein 3d structure reconstruction: accidental vs systematic errors over selected datasets

Identification of "hot spots" in protein structure: perturbation & (laplacian spectral) approaches

Identification of chromosomal aberrations in nuclear DNA

Characterization of long-range features (e.g. non-uniform nucleotide distribution in space)

Generation of Hi-c synthetic data (spectral methods)

Hi-c multiplex analysis (multiple cell types available)

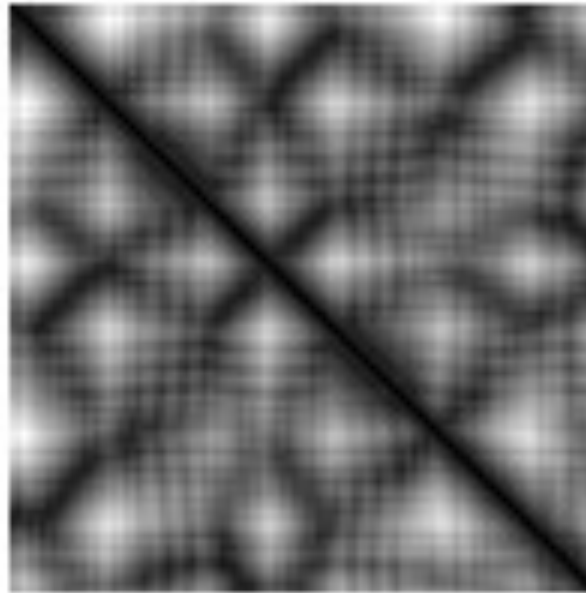
3D protein structure representation via contact maps

3D structure



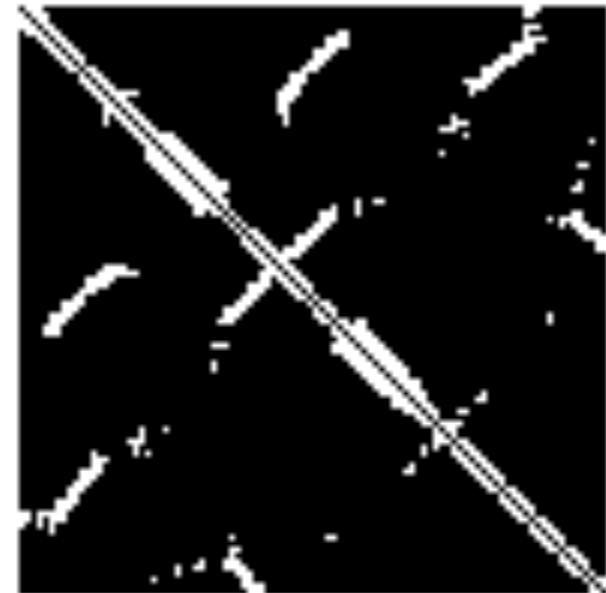
Get the 3D coordinates
of the residues

Distance matrix



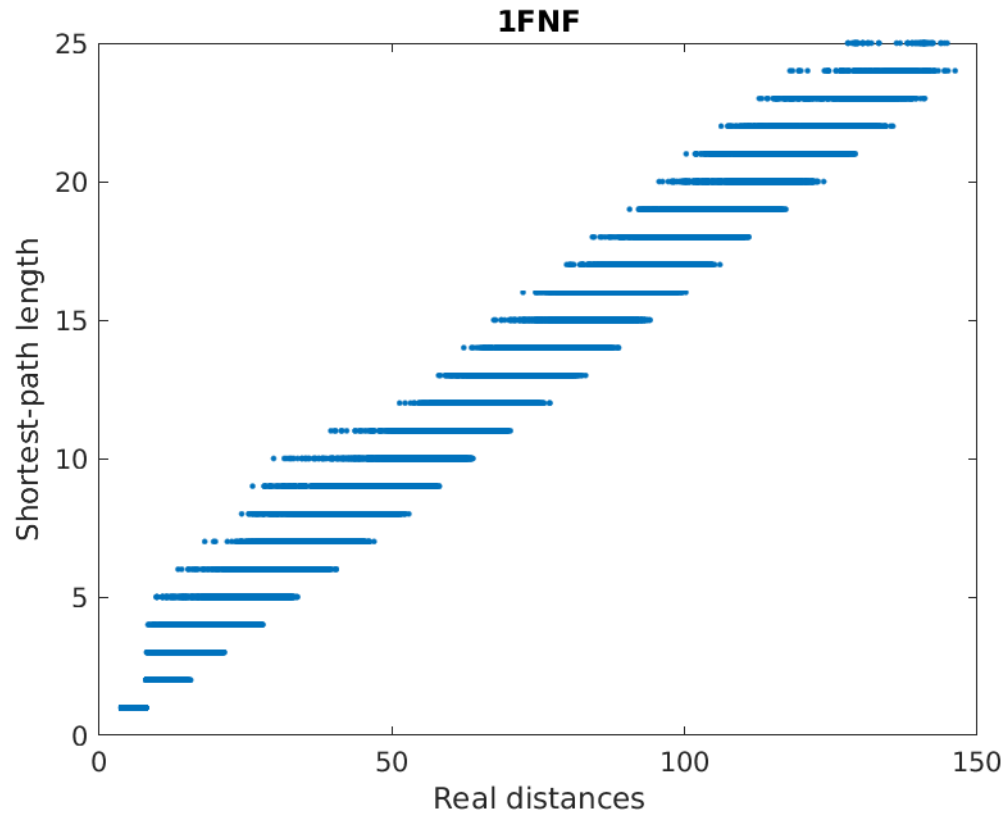
Compute the distance
between residues

Contact map

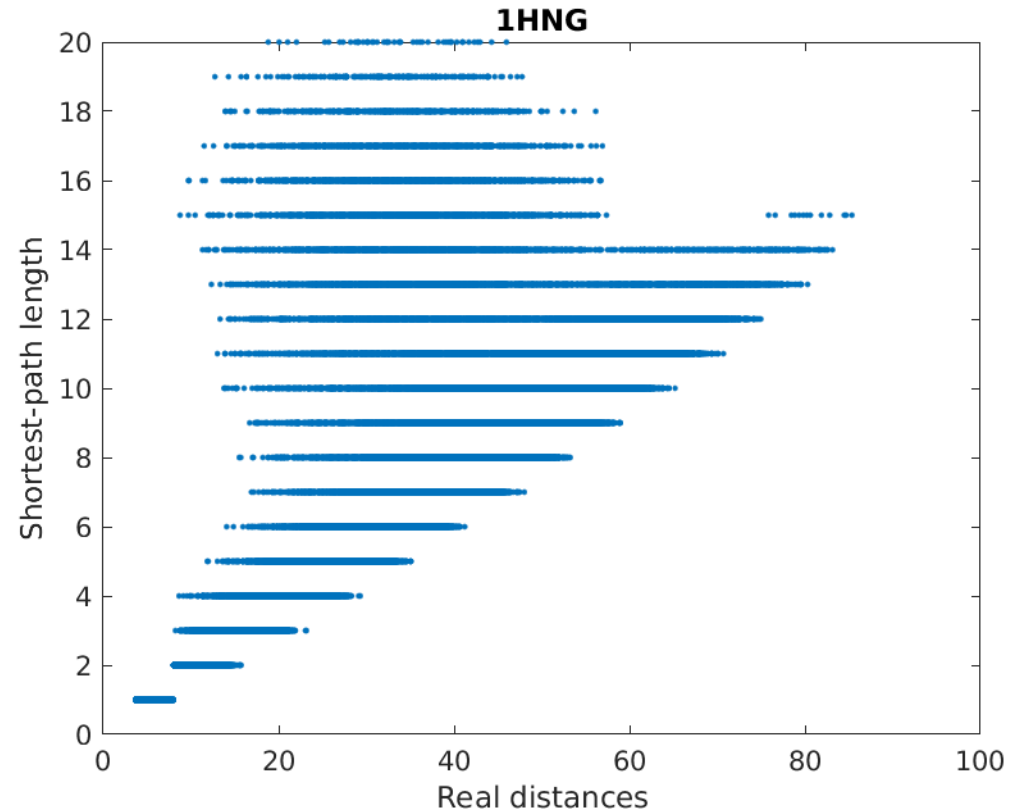


Backbone: thresholding the
distance matrix at 8-12 Å

ShRec3D reconstruction on protein contact maps



Correlation between real and reconstructed distances = **0.99**



Correlation between real and reconstructed distances = **0.68**

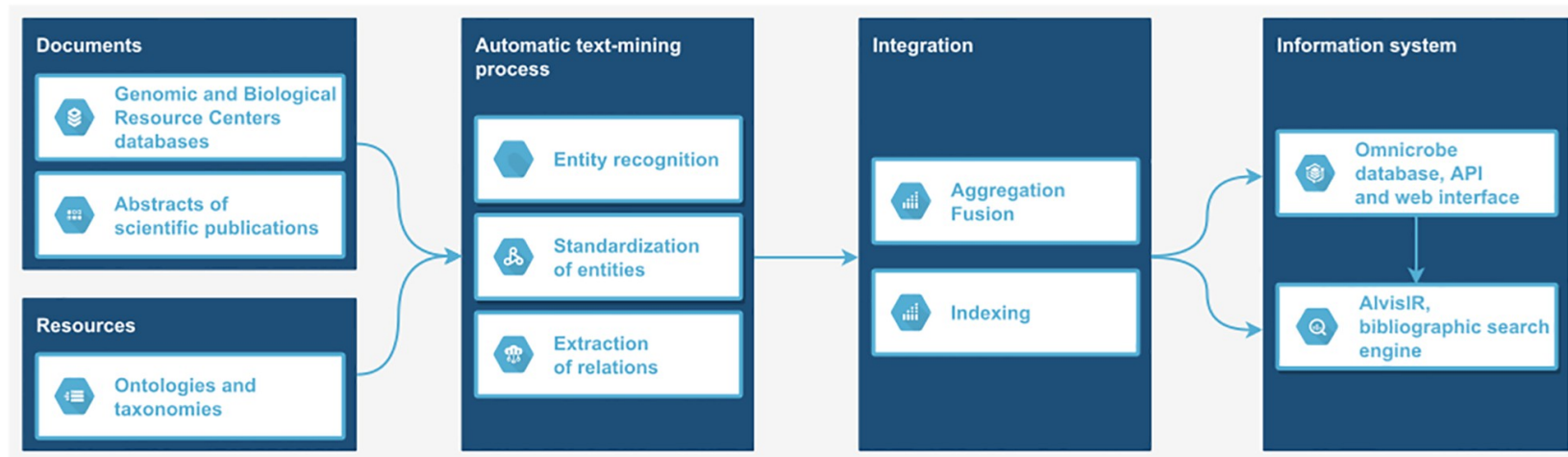
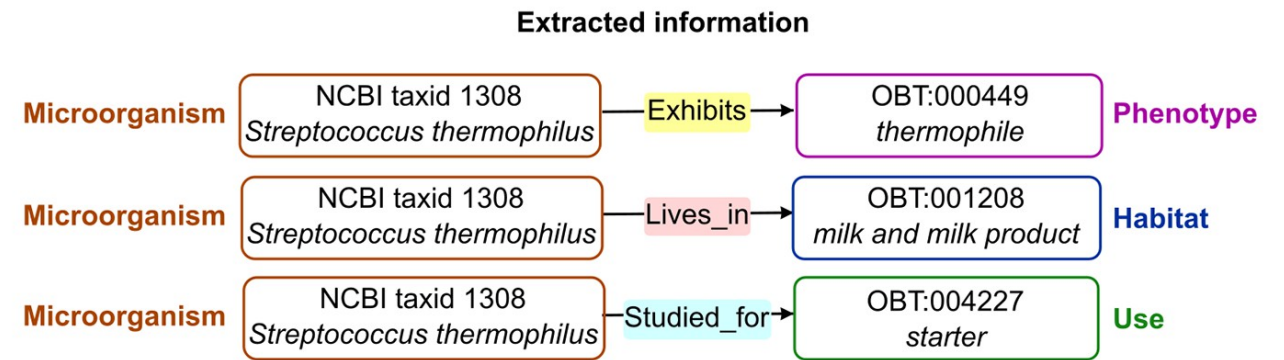
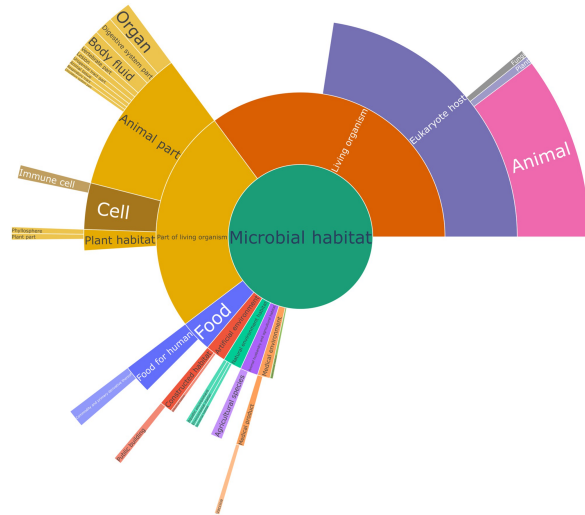
Hi-C & protein related projects

- Modularity: community, construction rules
- Aberration identification: node2vec, other
- Generation of synthetic hi-c data: spectral approaches
- Multiplex: compare different cell lines
- 3d reconstruction comparison with real structure: numerical and with VR
- Large scale characterization of proteins: rcsb.org + PFAM (pfam.xfam.org)

Knowledge graphs (FAIRomics project)

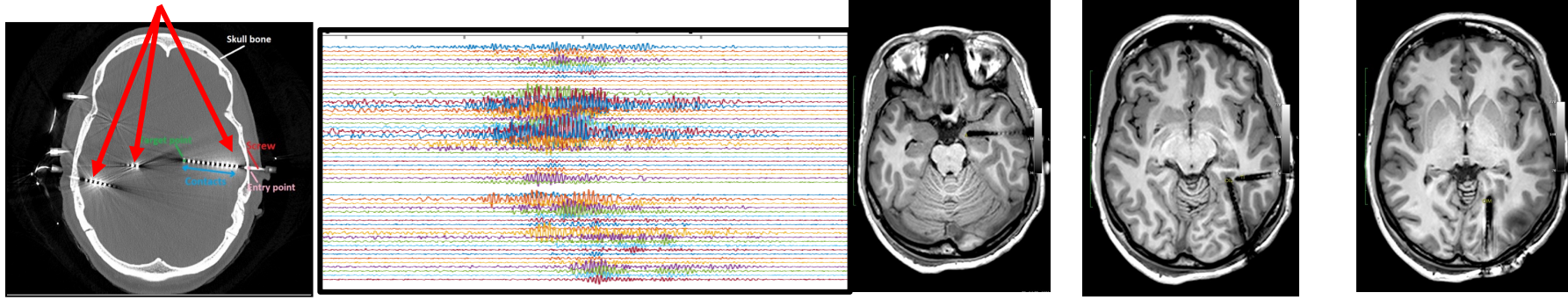
Reconstruction of Knowledge Graph
(relational graph, bipartite-like)

Analysis of existing knowledge graph
(eg Omnicrobe): communities,
centralities, inconsistencies, outliers,
relation prediction



Brain EEG data & epilepsy

Intracranial recordings from epileptic patients (sEEG): spontaneous and evoked activity



multi-scale brain network phenomenon:
identification of epileptogenic nodes
identification of epileptic episodes over time
high-order interactions in EEG recording channel
network

